

TB Limiter

DETAILED ENGLISH USER MANUAL

A stereo-linked true-peak mastering limiter with independent Input and Ceiling, micro-peak enhancement, program-aware RMS assist, balance correction, and TPDF dither.



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Host-visible endpoints documented: 10

1. Product purpose

A stereo-linked true-peak mastering limiter with independent Input and Ceiling, micro-peak enhancement, program-aware RMS assist, balance correction, and TPDF dither.

Final limiting needs to control inter-sample peaks without flattening musical recovery or shifting the stereo image. TB Limiter separates drive, ceiling, attack/release behavior, RMS-informed assistance, and optional balance correction so loudness and safety decisions remain explicit.

What result it creates

- Predictable true-peak ceiling control
- Stereo image stability
- Program-aware recovery and density
- Optional final 16- or 24-bit dither

Signal flow

**Input Gain -> fixed high-quality oversampling and lookahead -> peak/RMS-assisted gain computer
-> Enhance and RMS Balance behavior -> Ceiling -> optional TPDF Dither**

2. Complete feature guide

Input Gain and Ceiling

Input Gain drives material into limiting. Ceiling defines the independent final maximum. Raising Input changes loudness and reduction; changing Ceiling changes delivery headroom.

Attack and Release

Attack determines how the fixed lookahead shapes future peaks. Release sets recovery from 20 to 500 ms. Short release is dense; long release is smoother but may reduce movement.

Enhance

Adds controlled micro-peak shaping/density. Use after basic limiting is stable because it changes transient character.

RMS Assist

Adds program-aware RMS context to recovery so sustained energy and short peaks are not treated identically.

RMS Balance

Gently corrects persistent left/right energy drift while retaining stereo-linked peak limiting. Use only when a real long-term imbalance exists.

Dither

Off, 16-bit, or 24-bit TPDF dither is intended for the final word-length reduction. Do not add dither repeatedly at intermediate stages.

Programs

Programs include Transparent, Punch, Dense Control, Balance Guard, Maximize, Mastering, and RMS Smooth starting points.

3. Quick start

1. Set Ceiling for the delivery target.
2. Raise Input Gain until the required loudness or gain reduction is reached.
3. Tune Release for recovery and Attack for transient shape.
4. Add RMS Assist if sustained material pumps or peak-only behavior feels too nervous.
5. Use Enhance and RMS Balance only for a specific need.
6. Select Dither only on the final export stage.
7. Compare loudness-matched and check true-peak after encoding when relevant.

4. Practical workflows

Transparent master

1. Set Ceiling with codec headroom.
2. Use moderate Input Gain.
3. Keep Enhance and RMS Balance low or off.
4. Use medium Release and moderate RMS Assist.

Expected result: Peaks are controlled with minimal audible character.

Dense loud master

1. Increase Input Gain gradually.
2. Shorten Release until density rises without obvious chatter.
3. Add Enhance carefully.
4. Recheck low-frequency distortion and stereo stability.

Expected result: The master becomes louder and denser while remaining under the chosen ceiling.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- A limiter cannot repair clipping already printed into the source.
- True-peak compliance should be rechecked after lossy encoding.

- Dither belongs at the last bit-depth reduction only.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 10 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Attack VST3 ID 740224584	0.10 to 2.00	2.00	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Ceiling VST3 ID 660387005	-3.000 to 0.000	-1.000	Automatable	Sets or limits the final output level after the plug-in's main processing.
Dither VST3 ID 815960102	Off, 16-bit, 24-bit	Off	Automatable	Host-automatable control for Dither. Its full range and default are shown in this table; use it with the related feature section above.
Enhance VST3 ID 544326318	0.0 to 100.0	0.0	Automatable	Controls or enables non-linear density, harmonic color, or peak shaping.
Input Gain VST3 ID 1706566249	0.00 to 18.00	4.00	Automatable	Sets the level entering the processor and therefore changes downstream drive or detection.
Program VST3 ID 1886548852	Init, Transparent, Punch, Dense Control, Balance Guard, Maximize, Simple Limiting, Mastering, RMS Smooth	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Release VST3 ID 1090594823	20.0 to 500.0	120.0	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.
RMS Assist VST3 ID 2050725409	0.0 to 100.0	65.0	Automatable	Host-automatable control for RMS Assist. Its full range and default are shown in this table; use it with the related feature section above.
RMS Balance VST3 ID 1660932644	0.0 to 100.0	0.0	Automatable	Host-automatable control for RMS Balance. Its full range and default are shown in this table; use it with the related feature section above.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.